

(Solum)

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Software Requirements Specification Document

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1 Purpose

The rise of obesity has become one of the world's biggest issues in recent times. Almost 2 billion adults were overweight and 650 million were obese in 2016. The number of adults with obesity nearly tripled from over 40 years prior.¹ This goes in parallel with many people having a phone, in which 98% of people in the US are smartphone owners.² Many of these phone users are using digital applications, with over 136 billion apps downloaded in 2025³, with over 40% being health and fitness apps⁴. There's a significant market for the use of digital health and fitness apps in this age with the rise of app usage and obesity. In a study conducted among over 550 adults, 59.9% used for health and 38.2% used for fitness⁵. Even though there are existing applications, oftentimes they fail to cater to users' individual needs in a visual and comprehensive system. Meaning, there is an opportunity to create a new one that would resolve those shortcomings. Most digital fitness trackers specialize in a single aspect of: wellness, steps, calories, strength training, etc. This forces users to download multiple applications in order to cover all facets. Many fitness applications also have limited adaptability and often provide generic recommendations that don't adapt to personal goals and those with ones are locked behind a paywall, which drives users away. They also use a single metric such as weight or step count to calculate progress. This doesn't provide accurate representation of any progress made and can discourage users who look at such a one-dimensional criteria.

2 Scope

Solum is a standalone Java desktop fitness tracking application designed to help users organize workout activity, manage personal fitness information, and monitor workout progress within a centralized system. The application consists of account registration and login functionality, profile management, workout logging, workout history tracking, and a visualization of workout statistics. Users will be able to create an account, initializing it with information like height, weight, and fitness goal. The system supports both the Imperial and Metric system of measurement for display. Solum allows users to log workouts made up of multiple exercise entries, described with exercise name, sets, repetitions, and duration, and calculates the estimated calories burned. Workout sessions have a notes section for further personal tracking and organization. The application has a main dashboard that summarizes workout activity with categories such as total workouts completed, total calories burned, total workout minutes, and the latest workout calorie statistic. Users are able to review previous logged workouts through a workout history interface and delete records of selected workouts or entire profiles through a confirmation-based deletion. Solum has a workout statistics module that makes a visual diagram of caloric trends using a chart-based analysis. After three workouts have been logged, users are granted access to the personal plan feature which generates a personalized calorie projection and workout recommendations based on the user's selected goal and recent workout trends. Depending on the user's goal, the plan will adjust its target calorie projection and recommended calorie range for the following workout. The application utilizes

¹ <https://www.who.int/news-room/facts-in-pictures/detail/6-facts-on-obesity>

² https://www.pewresearch.org/internet/fact-sheet/mobile/?cb_viewport=tablet

³ <https://www.itransition.com/services/application/development/mobile/statistics>

⁴ <https://www.frontiersin.org/journals/public-health/articles/10.3389/fpubh.2022.914433/full#B13>

⁵ <https://pmc.ncbi.nlm.nih.gov/articles/PMC9931267/>

Java Swing for the graphical user interface and SQLite for persistent local database storage. The current version operates entirely as a local desktop application and does not require internet connectivity or external APIs for standard functionality. Features such as nutrition tracking, online synchronization, external profile sharing, and mobile deployment are outside the scope of the current implementation and may be considered for future development.

3 User characteristics

3.1 Key users

-College students

College students have difficulty balancing their time studying, spending time with their friends and family, but also needing to worry about their health, they are one of the key users who will benefit from being able to track physical activity, maintain health goals, and set fitness goals in order for them to be able to monitor their lifestyle habits. Their knowledge of the business is that of a novice as they have limited knowledge of health management and a centralized app will help them with guidance, and as for technological experience, college students are in the range of journeyman to a master as they are highly familiar with computers and desktop applications. Other characteristics that will have effect on the requirements would be that their age group would be primarily between the ages of 18-25, their attitude towards technology is that of positivity since they are a frequent user, and there are many that have physical disabilities that might restrict the uses of the app.

-Office workers

Office workers spend more than half of their workday sitting at their desk in order to reach the quota for the day. What they want is to be able to track their movement for the day, receive reminders for physical activity/breaks away from the desk to get up and stand, and monitor long-term health trends that are related to long periods of sitting. In terms of experience with technology and the subject, they are journeymen as they have a basic understanding of both concepts but are comfortable with what they already know. Due to prolonged periods of a sedentary lifestyle, office workers may experience back pain and other repetitive strain that may limit their physical activities and mobility. The age group is between 25-60 years old, so their views on technology would generally range between neutral and positive as they value the usefulness of digital fitness and health tracking applications.

-Overweight individuals

Those who are overweight who would want to use this product want to be able to track their physical activity, weight changes, and monitor their progress towards their weight related health goals. Overweight individuals are between that of a novice and a journeyman as these users generally understand the concept of weight management but may lack experience and knowledge to be able to know what to do to reach their goals. These key users would also have the experience of a journeyman in terms of technology as many are familiar with computers and digital fitness applications that can track their health but may not have the advanced knowledge to make use of other health features. Other characteristics that overweight individuals have that will affect the requirements and design of the product would be their physical abilities as the app should include low to moderate levels of physical exercise, and their intellectual abilities should govern the app to be more beginner friendly and intuitive in order for the information to reach their goals be more clearly presented.

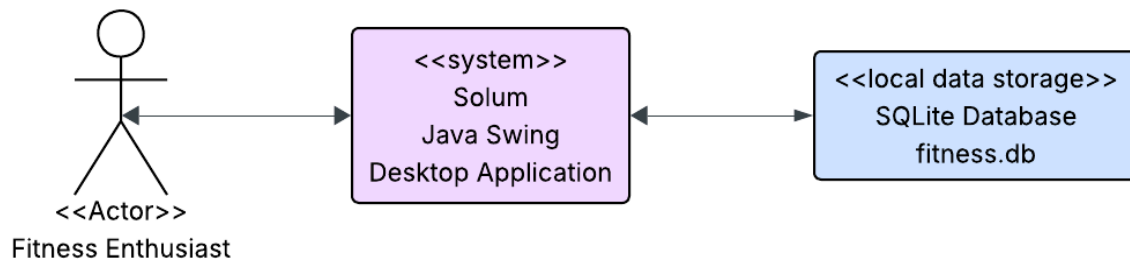
3.2 Secondary users

Doctors/Coaches/Trainers:

The current version of Solum does not provide external profile viewing, shared accounts, coach portals, doctor portals, or read-only access for outside users. Therefore, doctors, coaches, trainers, or other advisors are not direct users of the implemented system. However, these groups may still be considered indirect stakeholders. A user may choose to manually share workout history, calorie trends, or profile information from Solum with a trainer, coach, or healthcare professional outside of the application. Or the professional is able to track their patient/customer for them. In future versions, Solum may support read-only profile sharing or external review features, but this functionality is outside the scope of the current implementation.

4 Product perspective

4.1 System Context



4.2 User interfaces

- The system shall provide a graphical desktop interface using Java Swing.
- The system shall provide separate screens for login, registration, dashboard navigation, profile management, workout logging, workout history, and workout statistics.
- The system shall allow users to enter workout information including exercise name, sets, repetitions, duration, and workout notes.
- The system shall display workout statistics and calorie projections through graphical chart visualizations.
- The system shall display confirmation and error messages for invalid input, failed operations, successful saves, workout deletion, and profile deletion.
- The system shall provide navigation controls allowing users to move between dashboard, profile, workout logging, workout history, and statistics screens.
- The system shall support scrolling interfaces for viewing large workout history records and personalized workout plans.
- The system shall support both Imperial and Metric display formats for profile measurements.
- The system shall visually organize workout summaries, statistics, and workout history information through formatted interface panels.

4.3 Software interfaces

- The system shall use SQLite as the local database management system.
- The system shall use SQLite JDBC for communication between the application and the database.
- The system shall use Java Swing for graphical user interface rendering and user interaction.

- The system shall use JFreeChart and JCommon libraries for workout statistics visualization and chart rendering.
- The system shall initialize the SQLite database automatically during application startup.
- The system shall store user accounts, workouts, exercise entries, workout notes, calorie estimates, and profile preferences within the SQLite database.
- The system shall operate without requiring communication with external APIs, cloud systems, or third-party authentication services.

4.4 Deployment requirements

- The system shall be deployed as a standalone Java desktop application.
- The system shall require a Java Runtime Environment (JRE) or Java Development Kit (JDK) for execution.
- The system shall require the SQLite JDBC library for database connectivity.
- The system shall require the JFreeChart and JCommon libraries for workout statistics visualization.
- The system shall require the Orbitron font asset files used by the graphical user interface theme.
- The system shall create or initialize the local SQLite database during application startup if the database does not already exist.
- The system shall operate locally on the user's computer without requiring cloud infrastructure, external servers, or internet connectivity for standard operation.
- The system shall support persistent local storage of user accounts, profile information, workouts, exercise entries, workout notes, and statistics data.

5 Assumptions and Dependencies

- Users should have access to a computer capable of running a Java desktop application.
- Users should have Java installed or available through a Java development environment.
- Users should have the required project libraries included within the application setup (SQLite JDBC, JFreeChart, and JCommon).
- Users should have permission to create and modify local database files on their computer.
- Users should have a basic understanding of using desktop application interfaces.
- Users should be able to understand English interface labels and error messages.
- Users are responsible for needed hydration, warm-ups, and recovery activities before and after workouts.
- Users are responsible for receiving professional medical advice before beginning exercise programs.
- The application depends on SQLite for persistent local data storage.
- The application depends on Java Swing for graphical user interface rendering.
- The application depends on JFreeChart for workout statistics visualization.

6 Specific requirements

6.1 System Functional Requirements

R.1 Account Registration and Login

R.1.1 As a user, I want to create an account with a username and password so that I can securely store workout and profile information.

R.1.2 As a user, I want to enter my height, weight, and fitness goal during registration so that the system can personalize workout tracking and recommendations.

R.1.3 As a user, I want the system to validate account information during registration so that invalid profile data cannot be stored.

R.1.4 As a user, I want to log into the application using my account credentials so that I can access my saved workouts and statistics.

R.2 Profile Management

R.2.1 As a user, I want to view my profile information so that I can review my stored fitness data and preferences.

R.2.2 As a user, I want to edit my height, weight, and fitness goal so that my profile information remains accurate.

R.2.3 As a user, I want to switch between Imperial and Metric display units so that I can view measurements in my preferred format.

R.2.4 As a user, I want to delete my account profile so that I can permanently remove my stored data and workout history.

R.3 Workout Logging

R.3.1 As a user, I want to create workout sessions containing multiple exercise entries so that I can track complete workouts.

R.3.2 As a user, I want to enter the exercise name, sets, repetitions, duration, and workout notes so that detailed workout information can be recorded.

R.3.3 As a user, I want the system to estimate calories burned during workouts so that I can monitor workout intensity and activity levels.

R.3.4 As a user, I want the system to require at least one exercise entry before finishing a workout so that incomplete workouts cannot be saved.

R.4 Workout History

R.4.1 As a user, I want to view previously logged workouts so that I can review workout history and progress over time.

R.4.2 As a user, I want workout history records to display workout duration, calorie totals, notes, and exercise information so that workout sessions are easy to review.

R.4.3 As a user, I want to delete workout records so that incorrect or unwanted workout data can be removed.

R.5 Dashboard

R.5.1 As a user, I want to view dashboard summaries so that I can quickly review workout totals and recent activity.

R.5.2 As a user, I want to view total workouts, total calories burned, total workout minutes, and latest workout calories so that I can monitor overall workout progress.

R.5.3 As a user, I want dashboard navigation buttons so that I can quickly access workout logging, workout history, profile management, and workout statistics screens.

R.6 Workout Statistics and Personalized Recommendations

R.6.1 As a user, I want to view chart-based workout statistics so that I can visualize calorie trends and workout consistency.

R.6.2 As a user, I want the system to generate personalized calorie projections so that I can receive workout recommendations based on my selected fitness goal.

R.6.3 As a user, I want personalized workout recommendations to unlock after multiple workouts have been logged so that projections are based on sufficient workout history.

R.6.4 As a user, I want the system to display projected calorie ranges and workout recommendations so that I can better plan future workouts.

6.2 Logical Database Requirements

R.1 Data Entities and Attributes

R.1.1 User

The system shall store user account and profile information.

Primary Key: user_id

Attributes:

- *username (unique, required)*
- *password (required)*
- *height (required)*
- *weight (required)*
- *goal (required)*
- *unit_preference (required)*

R.1.2 Workout

The system shall store workout session information associated with a user.

Primary Key: workout_id

Attributes:

- *username (required)*
- *workout_datetime (required)*
- *duration (required)*
- *calories_burned (required)*
- *notes (optional)*

R.1.3 ExerciseEntry

The system shall store exercise entry information associated with a workout session.

Primary Key: exercise_entry_id

Attributes:

- *workout_id (foreign key reference, required)*
- *exercise_name (required)*

- *sets (required)*
- *reps (required)*
- *duration (required)*
- *calories_burned (required)*

R.2 Entity Relationships

R.2.1 User–Workout

Each user may be associated with zero or more workout records.

Each workout record shall be associated with the username of the user who created it.

R.2.2 Workout–ExerciseEntry

Each workout may be associated with one or more exercise entry records.

Each exercise entry shall be associated with exactly one workout.

R.2.3 Profile–Workout Data

A user profile shall store the user’s height, weight, selected fitness goal, and unit preference.

Workout and exercise records shall be used to generate workout history, dashboard summaries, calorie totals, and workout statistics.

R.3 Integrity Constraints

R.3.1 Uniqueness Constraints

Username shall be unique.

R.3.2 Required Data Constraints

User username shall not be null.

User password shall not be null.

User height shall not be null.

User weight shall not be null.

User goal shall not be null.

User unit preference shall not be null.

Workout username shall not be null.

Workout date and time shall not be null.

Workout duration shall not be null.

Workout calories burned shall not be null.

Exercise entry workout ID shall not be null.

Exercise entry name shall not be null.

Exercise entry sets shall not be null.

Exercise entry reps shall not be null.

Exercise entry duration shall not be null.

Exercise entry calories burned shall not be null.

R.3.3 Domain Constraints

Username shall be between 3 and 20 characters.

Password shall be between 4 and 30 characters.

Height shall be between 36 and 96 inches.

Weight shall be between 70 and 400 pounds.

Sets shall be between 1 and 20.

Repetitions shall be between 1 and 100.

Duration shall be between 1 and 300 minutes.

Exercise name shall be between 1 and 50 characters.

Goal shall be one of the supported goal values: WEIGHT_LOSS, WEIGHT_GAIN, or MAINTENANCE.

Unit preference shall be one of the supported unit values: IMPERIAL or METRIC.

R.3.4 Referential Integrity Constraints

ExerciseEntry workout_id shall reference an existing Workout workout_id.

Workout records shall be associated with the username of the account that created them.

R.3.5 Deletion Rules

If a user profile is deleted, the system shall delete the associated workout records and exercise entry records.

If a workout is deleted, the system shall delete the associated exercise entry records.

The system shall require confirmation before deleting a workout or user profile.

6.3 Software System Attributes

6.3.1 Usability

R.1 User Registration Efficiency

R.1.1 The system shall allow the user to complete account registration and initialize within 3 minutes under normal usage conditions.

R.1.2 The system shall display validation messages when invalid registration data is entered.

R.2 Workout Logging Efficiency

R.2.1 The system shall allow users to log a workout containing exercise entries through the graphical user interface.

R.2.2 The system shall allow users to complete a basic workout entry in under 90 seconds under normal usage conditions.

R.2.3 The system shall allow users to add multiple exercise entries to a single workout session.

R.3 Ease of Use

R.3.1 The system shall provide clearly labeled navigation controls for dashboard, profile management, workout logging, workout history, and workout statistics.

R.3.2 The system shall provide visually organized interface layouts for workout information, charts, and workout history records.

R.3.3 The system shall allow first-time users to navigate major application features without requiring external technical assistance.

R.3.4 The system shall support both Imperial and Metric display formats for profile measurements.

R.4 Error Prevention and Validation

R.4.1 The system shall display validation messages when required input fields are left empty.

R.4.2 The system shall prevent invalid height, weight, sets, repetitions, and duration values from being submitted.

R.4.3 The system shall prevent workout completion until at least one exercise entry has been added.

R.4.4 The system shall display confirmation dialogs before deleting workouts or user profiles.

6.3.2 Performance

R.1 Local Application Performance

R.1.1 The system shall process login requests within 2 seconds during normal local operation.

R.1.2 The system shall display dashboard summaries within 3 seconds during normal local operation.

R.1.3 The system shall retrieve workout history records within 3 seconds during normal local operation.

R.1.4 The system shall generate workout statistics charts and calorie projections within 3 seconds during normal local operation.

R.2 Workout Processing Capacity

R.2.1 The system shall support workout sessions containing multiple exercise entries.

R.2.2 The system shall support exercise duration values between 1 and 300 minutes.

R.2.3 The system shall support set values between 1 and 20.

R.2.4 The system shall support repetition values between 1 and 100.

R.2.5 The system shall support weight values between 70 and 400 pounds.

R.2.6 The system shall support height values between 36 and 96 inches.

R.3 Data Storage Performance

R.3.1 The system shall store workout and exercise entry information within the SQLite database during normal application operation.

R.3.2 The system shall maintain persistent local storage of user accounts, workouts, exercise entries, workout notes, and workout statistics after application restart.

R.3.3 The system shall support storage and retrieval of multiple workout records for each registered user.

6.3.3 Reliability/Dependability

R.1 Data Persistence

R.1.1 The system shall store user account, profile, workout, exercise entry, and statistics data within a persistent SQLite database.

R.1.2 The system shall preserve stored workout and profile information after normal application shutdown procedures.

R.1.3 The system shall allow previously stored workout and profile information to be retrieved after application restart.

R.2 Data Consistency

R.2.1 The system shall automatically update workout duration totals when exercise entries are added to a workout.

R.2.2 The system shall automatically update workout calorie totals when exercise entries are added to a workout.

R.2.3 The system shall maintain valid relationships between users, workouts, and exercise entries during database operations.

R.3 Recovery Protection

R.3.1 The system shall require confirmation before permanently deleting workout records.

R.3.2 The system shall require confirmation before permanently deleting user profile records.

R.3.3 The system shall prevent incomplete workout records from being finalized without exercise entries.

6.3.4 Security

R.1 User Authentication

R.1.1 The system shall require valid username and password credentials before allowing access to user profile and workout information.

R.1.2 The system shall prevent duplicate usernames during account registration.

R.1.3 The system shall deny login access when invalid account credentials are entered.

R.2 Data Validation Protection

R.2.1 The system shall validate height, weight, sets, repetitions, duration, username, and password values before storing data within the database.

R.2.2 The system shall prevent empty required input fields from being submitted.

R.2.3 The system shall prevent invalid numeric values from being stored within workout and profile records.

R.3 Workout and Profile Protection

R.3.1 The system shall allow users to delete only workout records associated with the currently logged-in account.

R.3.2 The system shall restrict profile modification operations to the currently logged-in user account.

R.3.3 The system shall require confirmation before permanently deleting workout records.

R.3.4 The system shall require confirmation before permanently deleting user profile records.

R.4 Local Data Privacy

R.4.1 The system shall store user profile and workout information locally within the SQLite database.

R.4.2 The system shall allow users to remove locally stored profile and workout information through deletion functionality.

6.3.5 Maintainability

R.1 Modular Architecture

R.1.1 The system shall separate graphical user interface components, controllers, database management logic, and workout management functionality into independent classes.

R.1.2 The system shall organize account management, profile management, workout logging, workout history, and workout statistics functionality into dedicated controller and manager classes.

R.1.3 The system shall maintain separate classes for user profiles, workouts, and exercise entry data models.

R.2 Database Maintainability

R.2.1 The system shall maintain workout, exercise entry, and user profile information within separate database tables.

R.2.2 The system shall maintain relationships between users, workouts, and exercise entries through structured database references.

R.2.3 The system shall allow workout and profile data to be modified without requiring recreation of the database.

R.3 Extensibility

R.3.1 The system shall support future expansion of workout statistics and personalized workout recommendation functionality.

R.3.2 The system shall support future addition of new workout-related features without requiring complete redesign of the application structure.

R.3.3 The system shall support future expansion of profile management and workout tracking functionality.

R.4 Interface Maintainability

R.4.1 The system shall maintain separate graphical interface screens for login, registration, dashboard, profile management, workout logging, workout history, and workout statistics.

R.4.2 The system shall allow updates to graphical user interface layouts without requiring modification of database storage logic.

R.4.3 The system shall maintain reusable graphical styling components for consistent interface appearance across application screens.